

Exploring Non-Traditional Health Data: Social Determinants of Health (SDOH)

Marvin Azuogu

Houston City College

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Professor: Dr. Doreen Rosenstrauch

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Introduction

For decades, medical data was strictly clinical, blood pressure readings, heart rates, and lab results. However, the modern healthcare landscape is shifting toward "non-traditional" health data to gain a more complete picture of patient wellness. One of the most significant categories of this data is the Social Determinants of Health (SDOH). These factors represent the conditions in which people are born, grow, live, work, and age, and they often have a greater impact on health outcomes than medical care itself.

Understanding Social Determinants of Health

Social determinants of health are the non-medical factors that influence health outcomes (WHO, 2024). They are the "upstream" drivers of wellbeing that occur outside of a doctor's office. While clinical care can treat a disease, SDOH often determines whether a person gets sick in the first place.

Three primary examples of SDOH include:

- **Economic Stability:** This involves a person's income, cost of living, and job security. Financial instability often leads to "food insecurity," where individuals cannot afford the nutritious meals required to prevent chronic illness (CDC, 2024).
- **Neighborhood and Built Environment:** This includes access to safe housing, clean water, and green spaces. For instance, living in an area with high air pollution or limited

transportation can prevent a patient from reaching follow-up appointments (Artiga & Hinton, 2018).

- **Education Access and Quality:** Higher levels of education are linked to better health literacy, higher income potential, and a longer life expectancy.

Improving Health Outcomes Through SDOH Data

Integrating SDOH data into healthcare allows for "precision medicine" that goes beyond biology. When a health system identifies that a patient has unstable housing, they can coordinate with social workers rather than simply prescribing a pill that requires refrigeration the patient doesn't have. Recent research suggests that utilizing SDOH data can significantly reduce hospital readmission rates (Guo et al., 2023). By identifying "hot spots" in a city where asthma rates are high due to poor housing conditions, public health officials can intervene at the environmental level rather than just treating individual patients one by one. This data allows for a more proactive, preventative approach to population health management.

Ethical Concerns and Potential Issues

Despite the benefits, using non-traditional health data raises serious ethical concerns. The first is privacy. Patients may feel that sharing information about their income, home life, or legal status is an invasion of privacy, especially if they fear that data could be used against them by insurance companies or employers (Magnan, 2021). Another concern is algorithmic bias. If health systems use AI to analyze SDOH data, the algorithms may inadvertently penalize marginalized communities. For example, if an algorithm labels a patient as "high risk" for non-compliance because they lack a car, it could lead to them being denied certain elective procedures (Guo et al.,

2023). There is also the risk of "medicalizing" social problems, treating poverty as a clinical diagnosis rather than a societal issue that requires policy-level change.

Conclusion

Non-traditional health data, specifically Social Determinants of Health, provides the "missing pieces" of the puzzle in modern medicine. While traditional clinical data tells us *what* is happening to a patient's body, SDOH data tells us *why* it is happening. In my view, the importance of this data cannot be overstated; we cannot achieve true health equity without addressing the social environments that dictate a person's ability to thrive. While we must remain vigilant about privacy and bias, integrating these social insights is the only way to build a healthcare system that is truly holistic and effective for everyone.

References

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